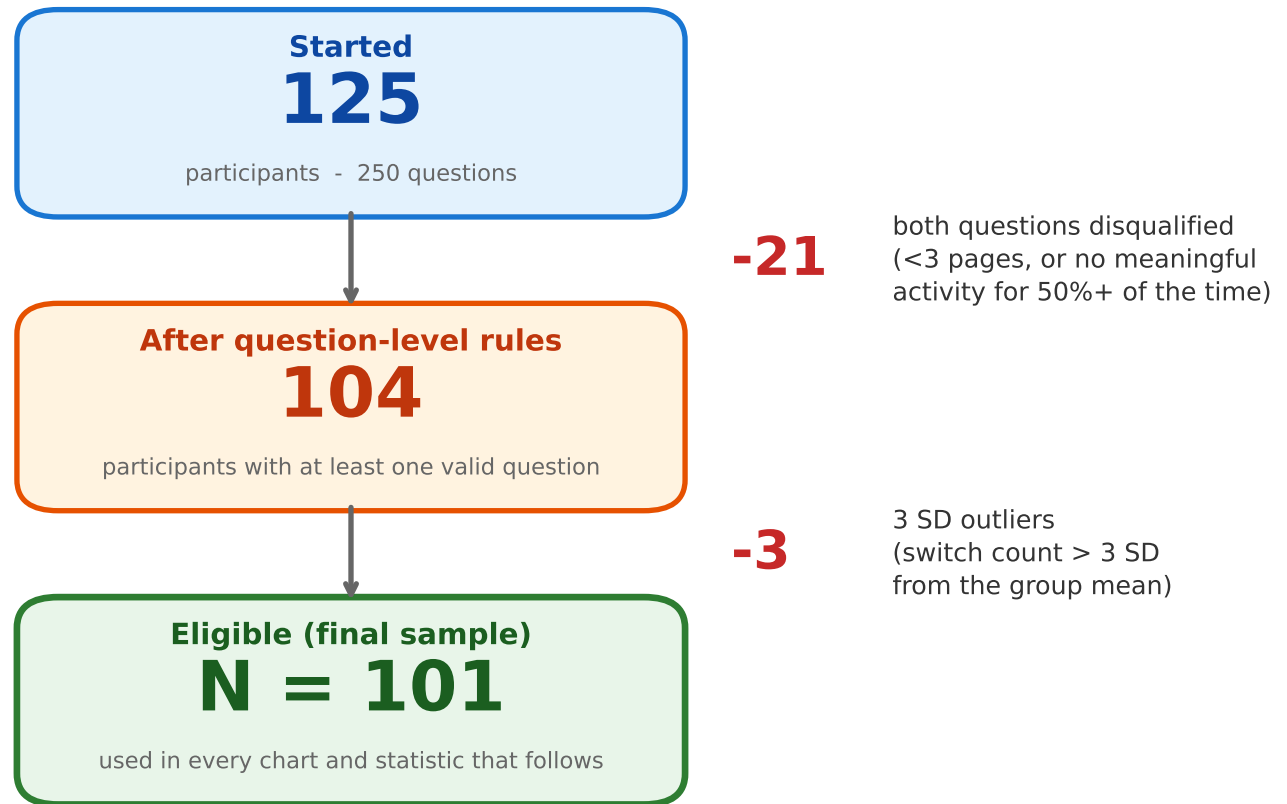
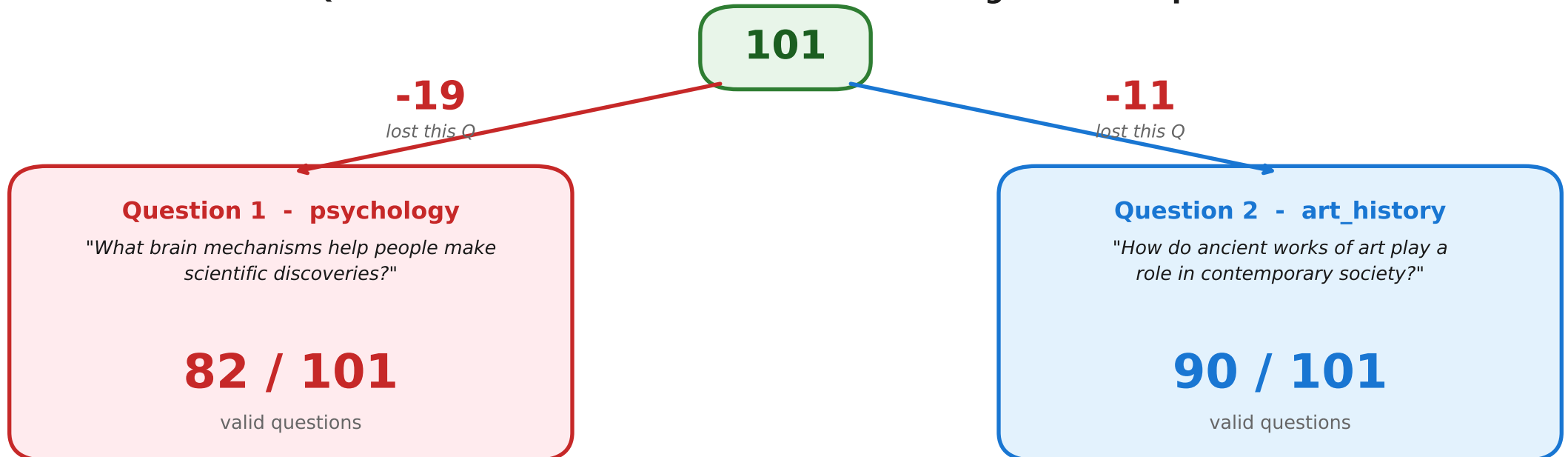


M55: Exclusion Audit

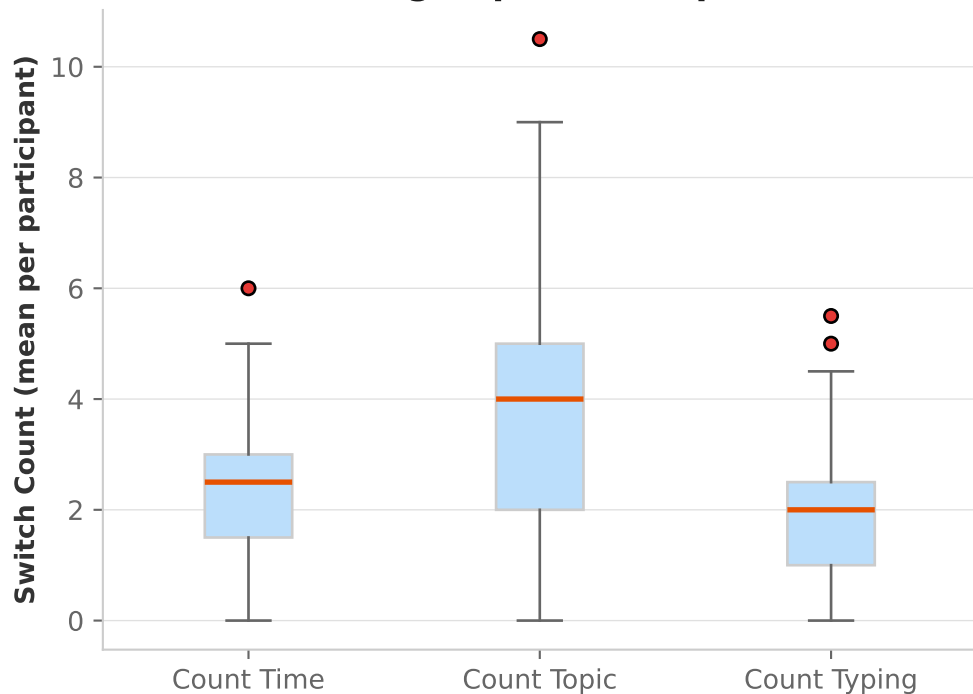


Question-Level Exclusion within the 101 Eligible Participants

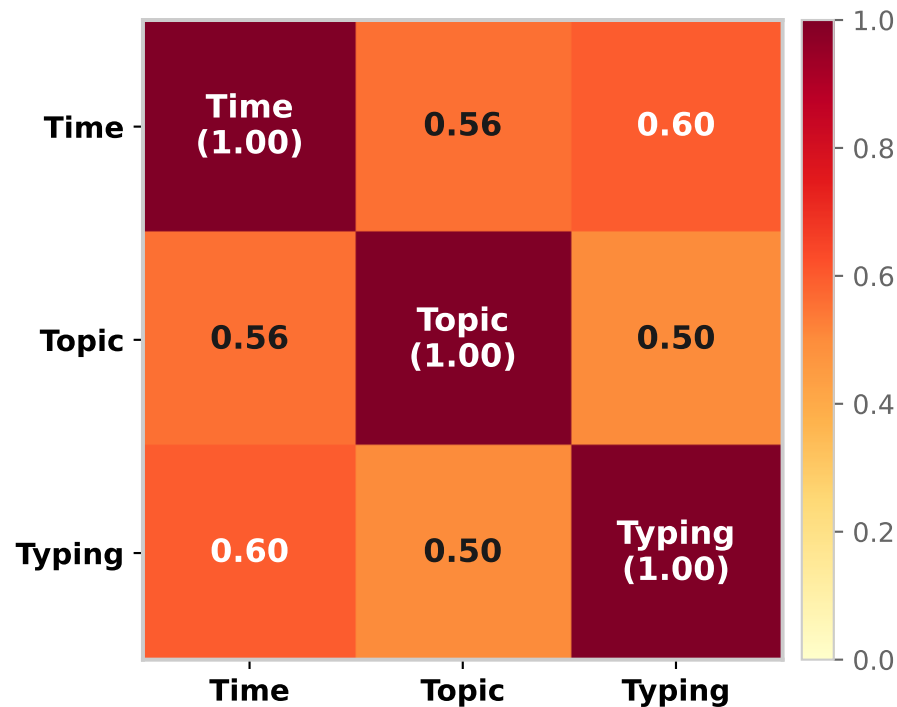


Descriptive Statistics: Raw Switch Counts

Distribution of Raw Switch Counts (Averaged per Participant)

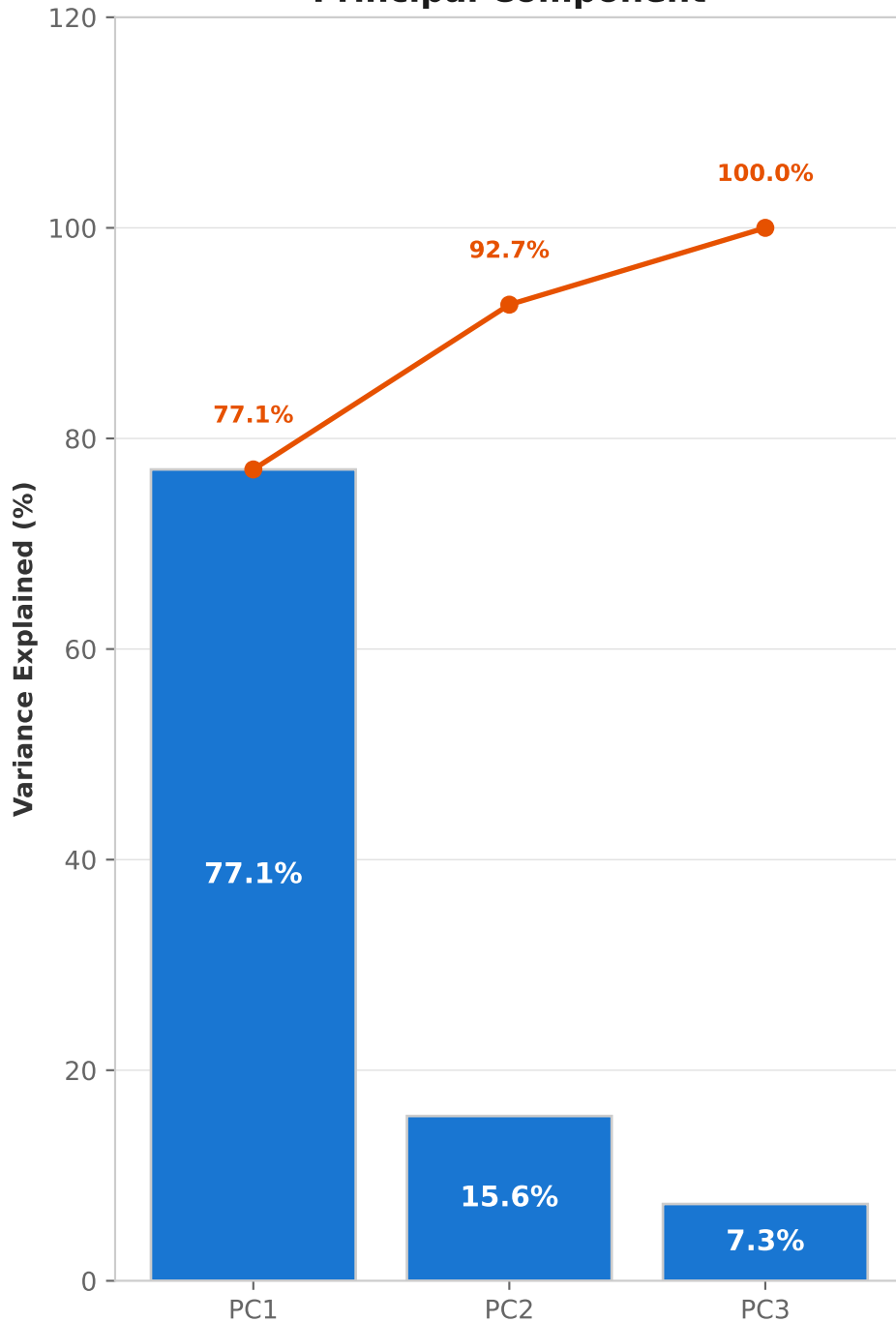


Correlation Between Switch Count Variables



PCA Results: Dimensionality Reduction on Raw Switch Counts

Variance Explained by Each Principal Component



PCA Loadings (variable weights per component):

PC1 (77.1% variance):

Count Topic	+0.879	+++++
Count Time	+0.374	+++++
Count Typing	+0.297	++++

PC2 (15.6% variance):

Count Time	+0.682	+++++
Count Typing	+0.554	+++++
Count Topic	-0.478	-----

PC3 (7.3% variance):

Count Typing	+0.778	+++++
Count Time	-0.629	-----
Count Topic	+0.004	

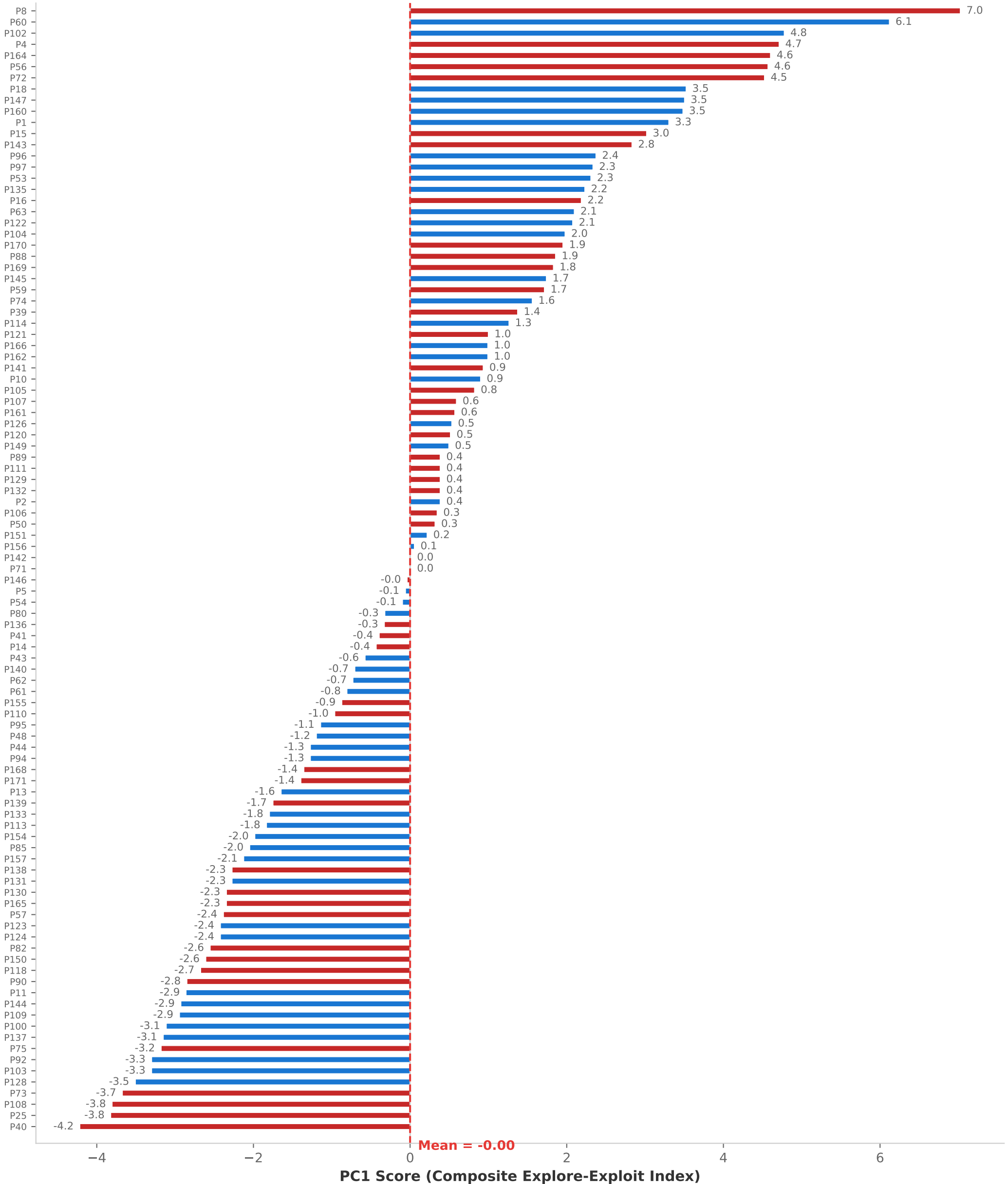
Interpretation:

PC1: Primarily reflects topic switching
(loading = +0.879)
Higher score = more topic transitions

PC2: Primarily reflects time switching
(loading = +0.682)
Higher score = more time transitions

PC3: Primarily reflects typing switching
(loading = +0.778)
Higher score = more typing transitions

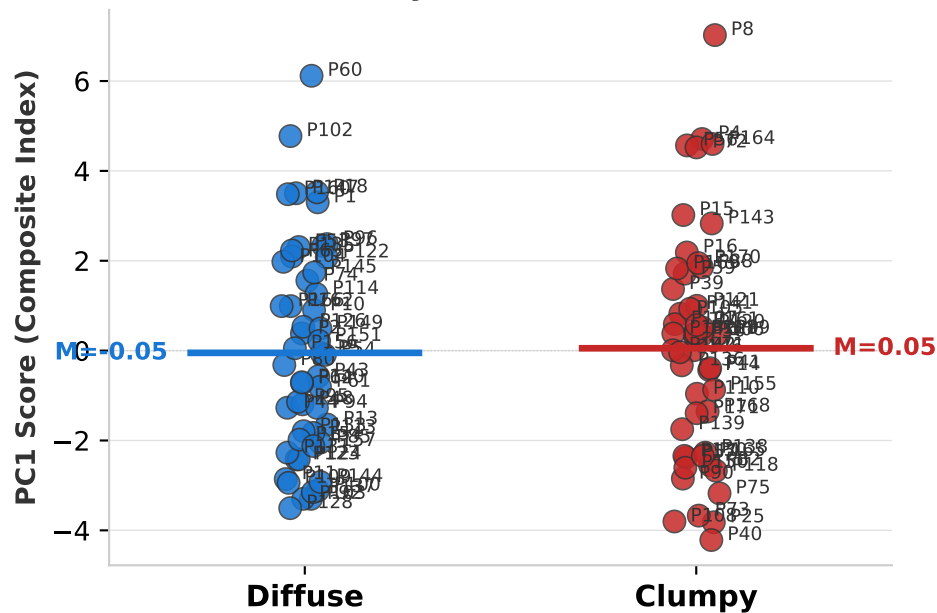
Individual PC1 Scores: Composite Explore-Exploit Index



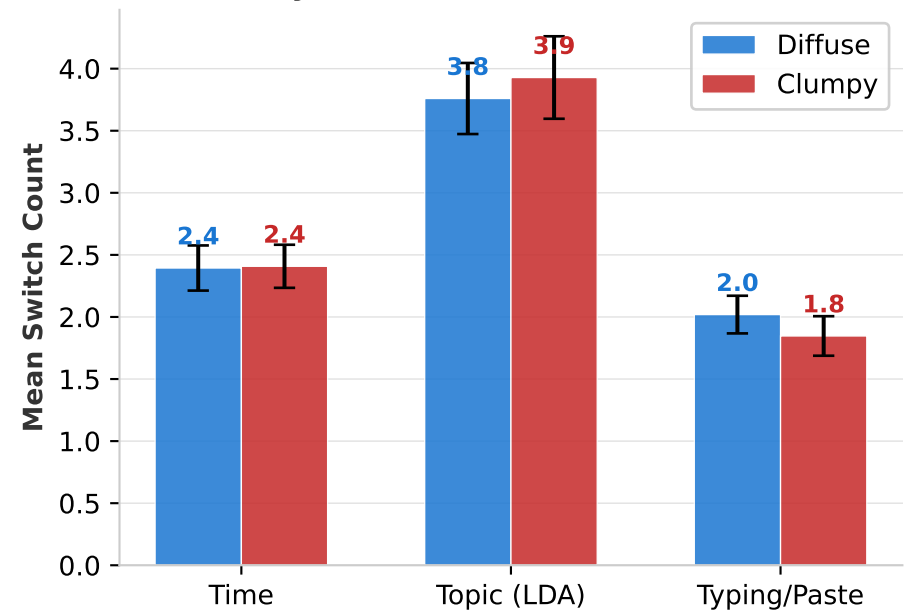
Condition Comparison: Clumpy vs Diffuse

N = 101 (Diffuse: 52, Clumpy: 49)

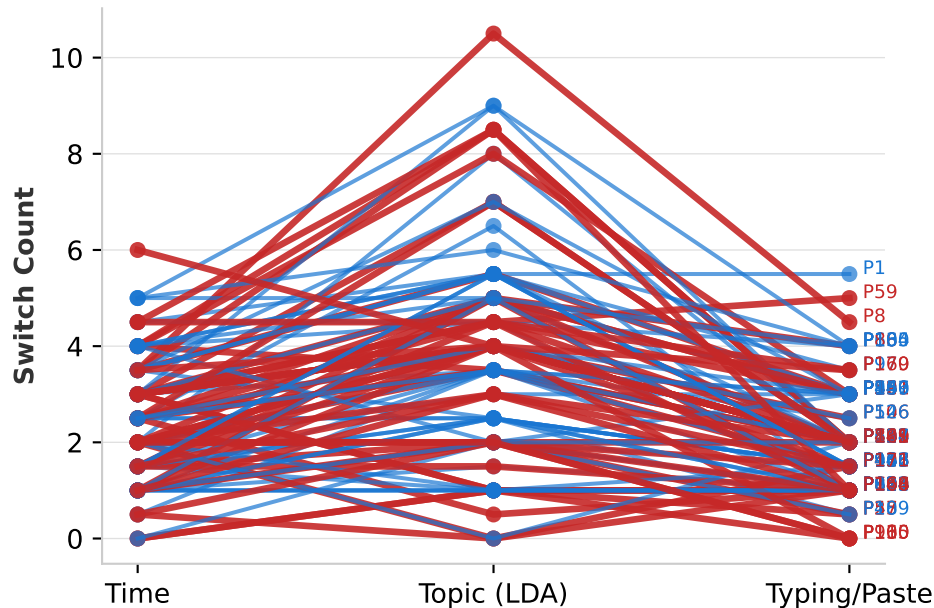
PC1: Composite Explore-Exploit
by Condition



Raw Switch Counts
by Condition (Mean $\pm$ SE)



Individual Profiles
by Switch Type



Condition Comparison: Descriptive Statistics & t-Tests

N = 101 (Diffuse: 52, Clumpy: 49)

DESCRIPTIVE STATISTICS (M / SD / SE per condition)

Variable	Diffuse (n=52)			Clumpy (n=49)		
Time	M= 2.39	SD=1.31	SE=0.18	M= 2.41	SD=1.21	SE=0.17
Topic (LDA)	M= 3.76	SD=2.06	SE=0.29	M= 3.93	SD=2.33	SE=0.33
Typing/Paste	M= 2.02	SD=1.09	SE=0.15	M= 1.85	SD=1.11	SE=0.16
PC1 (composite)	M=-0.05	SD=2.34	SE=0.32	M= 0.05	SD=2.52	SE=0.36

SE = SD / sqrt(n); n is post-exclusion non-NaN count.

INDEPENDENT-SAMPLES t-TESTS (Welch)

Variable	t	df	p	Cohen d	Sig.
Time	-0.06	99.0	0.9559	+0.01	n.s.
Topic (LDA)	-0.39	96.0	0.7008	+0.08	n.s.
Typing/Paste	0.78	98.4	0.4351	-0.16	n.s.
PC1 (composite)	-0.21	97.2	0.8329	+0.04	n.s.

* p < .05 ** p < .01 *** p < .001

INTERPRETATION

PC1 difference (Clumpy - Diffuse): +0.10
 Clumpy condition shows more explore-exploit switches
 Effect size (Cohen's d): +0.04
 Result is NOT statistically significant (p = 0.8329)
 Low statistical power due to small sample (N = 101)

Largest sub-variable effect: Typing/Paste
 d = -0.16, p = 0.4351

Note: Typing metric uses answer_snapshot bursts + paste events.